

Initial Assessment of Local Emergency Operation Centers in 32 Earthquake Affected Districts of Nepal



Report by:



Housing Recovery and
Reconstruction Platform

Primarily funded by:



2021

About HRRP

The Housing Recovery and Reconstruction Platform (HRRP) was established in December 2015 to take over supporting coordination of the post-earthquake housing reconstruction from the Nepal Shelter Cluster, as it returned to the pre-earthquake format as a standard cluster. HRRP provides coordination support services for the National Reconstruction Authority (NRA), Building and Grant Management and Local Infrastructure (GMALI) Central Level Project Implementation Units (CLPIUs), other relevant government authorities, and Partner Organisations (POs). HRRP is currently in its fourth phase and will continue its coordination services until July 2021. HRRP is primarily funded by UKAid Nepal and led by CRS Nepal with technical leadership from NSET.

The HRRP works at municipal, district, sub-national, and national levels, providing support primarily to the 14 districts most affected by the earthquake, as well as roving support to the 18 districts moderately affected by the earthquake.

Areas of Focus

The HRRP's main areas of focus are:

- Monitoring and documenting the housing reconstruction process.
- Facilitating coordination and experience sharing to support improvements in coverage and quality of socio-technical assistance.
- Supporting collective planning and roll out of additional support for vulnerable households.
- Addressing gaps and duplications.
- Advocacy, communications, and research.



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Acronyms

DAO	District Administration Office
DEOC	District Emergency Operations Center
DMIS	Disaster Management Information System
DRRM	Disaster Risk Reduction & Management
DRR	Disaster Risk Reduction
EOC	Emergency Operations Center
EWS	Early Warning System
HRRP	Housing Recovery and Reconstruction Platform
LDCRP	Local Disaster and Climate Resilience Planning
LEOC	Local Emergency Operations Center
LEPRP	Local Emergency Response Planning
PEOC	Provincial Emergency Operations Center
MEOC	Municipal Emergency Operations Center
MIRA	Multi-Cluster Initial Rapid Assessment
MoCIT	Ministry of Communication and Information Technology
MoHA	Ministry of Home Affairs
NEOC	National Emergency Operation Center
SFDRR	Sendai Framework for Disaster Risk Reduction
SOP	Standard Operating Procedure
VCA	Vulnerability Capacity Assessment

Executive Summary

Background

This report presents findings of a LEOC (Local Emergency Operation Centre) Assessment conducted by Housing Recovery and Reconstruction Platform (HRRP) at 282 local levels of 32 earthquake affected district to get initial status of LEOC so as to describe status of LEOC and its functionality at different local level. The assessment's findings also provide a strong base for coordination with governments at federal, provincial and local level for strengthening DRR including LEOC. A Cross-Sectional Study design approach was applied for the assessment that included a close-ended questionnaire for survey.

Despite the efforts to reduce limitations of the assessment, this report has to be read and referred to considering certain issues. In addition, LEOC Assessment is conducted at 282 local levels of 32 earthquake affected districts only because of which findings and recommendation of this assessment reflect the overall scenario of LEOC in Nepal. Data for this LEOC Assessment was collected remotely which was conducted from the beginning of first week of January 2021 until the second week of February 2021 as the field movement was restricted due to COVID-19 pandemic. Besides this, assessment did not cover recent initiatives on DRR undertaken by the local levels and might not provide detailed information on overall function of EOCs at local level.

Findings of the Study

Respondents for this assessment were DRR focal persons of the local governments, where DRR focal persons of the local levels were found from bureaucrats (84%) and elected bodies (16%), though the responses from government officials were high. According to DRR focal persons, landslides, flood (64%), earthquake (61%) and fire (50%) are the four major hazards at local levels, which are also common hazards across urban and rural municipalities. When DRR focal persons were asked to rate the vulnerability of their Palika, 69 respondents considered that their Palika is highly vulnerable 185 rated their Palika as having medium vulnerability, while 28 rated their Palika as low to disaster risk. This rating is based on the perception and few indicators like frequency and intensity of hazards. Findings of this LEOC Assessment further elucidates that local governments have given less priority for DRR planning as well as Emergency Response Planning. Only 103 local governments have developed Local Disaster and Climate Resilience Planning (LDCRP) and Local Emergency Preparedness and Response Plan (LEPRP) is developed by 112 local levels only. Similarly, provisions of Damage Need Assessment (DNA) were found at 100 local levels. Though landslides, flood, earthquake, and fire are found to be major and common hazards across all the local levels in 32 earthquake affected districts, efforts for establishment of EWS have not been initiated by local governments. Only 13 interviewed Palika have established EWS, where Flood Early Warning system is established at local levels like Bigu Rural Municipality, Gauri Shankar Rural Municipality, and Baiteshwor Rural Municipality of Dolakha, Ratnanagar Municipality of Chitwan, Gairidhar Municipality, and Kandasiri Municipality of Nawalparasi-East, Musikot Municipality and Gulmi Durbar Rural Municipality of Gulmi and in Tinau Rural Municipality of Palpa. Likewise, Dharche Municipality of Dhading, has developed a Landslide EWS. Respondents from Hetauda Sub-Metropolitan City of Makwanpur, Devchuli Municipality of Nawalparasi-East, and Bharatpur Metropolitan City of Chitwan informed that they have traffic signals as EWS.

The assessment found that LEOC is established in only 16 local levels of 32 earthquake affected districts, where LEOC is fully functional at 14 local levels, yet 24 hours × 7 days operation of LEOC is found in nine local levels (Palikas) only. Of the 16, separate offices for LEOC are found in 11 local levels whereas the nine LEOC are fully equipped with IT equipment, furniture, and life saving equipment.

SOPs, which are required for proper functioning of LEOC, are found at 11 local levels. However, of this, the SOPs made by six local levels clearly indicate reporting of hazards incident in an emergency and in normal situations. Likewise, only nine Palikas have been doing regular simulation exercises for strengthening the functioning capacity of LEOC. Nine EOCs at local level have established linkage with district, province, and federal governments, but none of the Palikas created linkage between EWS and LEOC. In the LEOC established Palikas, operational and technical challenges are the main challenges that hinders smooth functioning of LEOC. When asked about the reason behind not establishing LEOC, limited technical knowledge and inadequate funds were mentioned as the major reasons during the survey.

Conclusions & Recommendation

Overall, LEOC is absent in many local levels of 32 earthquake affected districts because of budget deficiency and limited technical expertise. Even though LEOC is established in some of the urban and rural municipalities, they are facing technical and operational challenges for proper functioning. Local Governments' leadership on DRR is substantial for disaster risk governance and promoting community resilience. Therefore, it is important that the local government should prioritize the establishment of LEOC taking consultation from federal and provincial governments including DRR practitioners. Local governments also need to focus on strengthening institutional capacity for making LEOC fully operational with clearly defined roles and responsibilities.



Chapter I: Introduction

1.1 Background

Located between 26°20' to 30°10' north latitude and 80°15' to 88°19' east longitude, Nepal shares a border with the People's Republic of China in the North and its eastern, southern and western borders with India. Nepal is not only renowned for its abundance of natural landscape but is also known as disaster prone countries in the world. Every year in Nepal different geological and hydro-meteorological disastrous incidents claim a larger number of lives and cause significant economic damage.

According to MoHA disaster reports, more than 80% of the population is exposed to the risk of natural hazards, which include earthquakes, droughts, floods, landslides, extreme temperature, and glacier lake outburst floods (GLOFs). Likewise, rugged and fragile geological structures with variable climatic conditions across different ecological zones are considered major causes of disaster in Nepal. However, in the present context, rapid and unplanned urbanization, increasing population and limited knowledge and understanding of disaster risk reduction have further intensified disaster risk in Nepal.

The DRRM Act 2017 (<https://bit.ly/2WWKZ9r>) stipulated Local Emergency Operation Center (LEOC) as a mandatory task to collect data, and to coordinate and cooperate with local level agencies for disaster preparedness and management. The DRRM Act 2017 has stipulated that each local government must establish and operate LEOC and develop an effective Disaster Management Information System (DMIS) in each local government. Similarly, based on the SFDRR (<https://bit.ly/3xpvXFC>), the Government of Nepal developed and endorsed Disaster Risk Reduction National Strategic Plan of Action (2018- 2030) (<https://bit.ly/3rVZTs3>), as Nepal's roadmap to implement the key provisions of SFDRR in context to Nepal. Of the four prioritized actions of SFDRR (i.e. Sendai framework), point two and four focus on “Strengthening Disaster Risk governance to manage disaster risk and Enhance disaster preparedness for effective response and to “Build Back Better” in recovery, rehabilitation, and reconstruction”. In accordance with priority action point two, disaster risk governance entails mechanisms, institutions and processes for effective DRRM at all levels. The priority action four of SFDRR covers disaster risk forecasting, early warning system, emergency preparedness plan, collection and storage of emergency items, safe evacuation, and mock drills enhancing disaster preparedness.

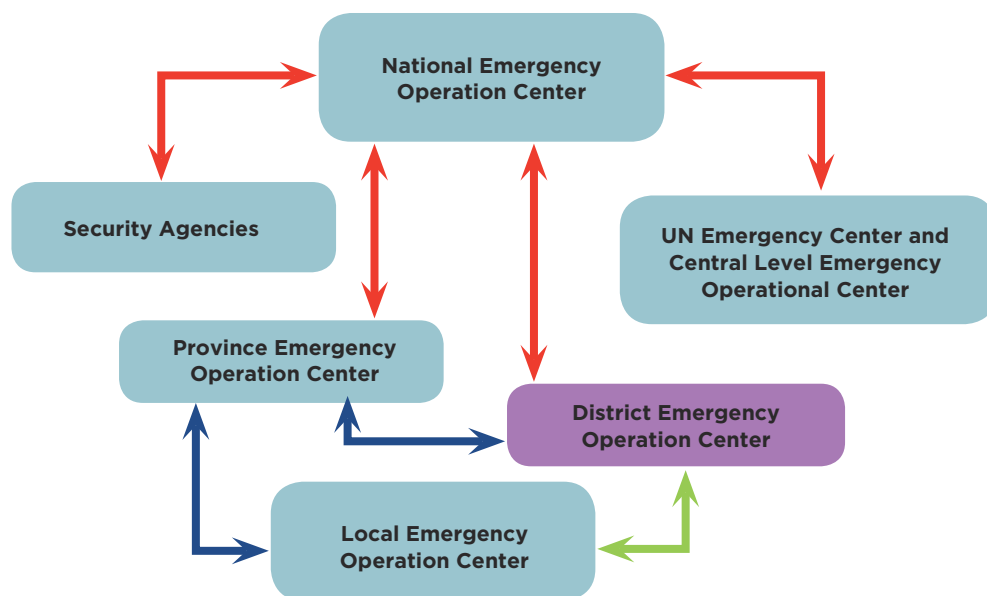


Figure 1: Structure of EOCs in Nepal (Source: Nepal Disaster Reduction Portal)

Based on these global goals and DRRM Act 2017, provision of LEOC is one of the milestones for addressing disaster risk as well as safeguarding life and assets of the people.

An EOC is the place where information is collected, evaluated, displayed, and all operations related to disasters are coordinated. Importance of EOC was first realized after the Koshi flood in 2008 in eastern Nepal, when one of the Army check posts informed about a burst in Koshi dam to the lower stream community of the then four village development committees (VDCs) of Sunsari district. Thereafter, the National Emergency Operation Center (NEOC) was established on 17 December 2010 located within the MoHA premises in Singhadurbar. The federal structure of the country has restructured EOCs function and working areas from top to bottom. Currently, four levels of EOC are in operation in Nepal including NEOC at the federal level, PEOCs at the provincial level, DEOCs at the district level and LEOCs at the local level. The Ministry of Home Affairs is the lead government agency of NEOC, DAO office is the lead government agency of DEOC, and Municipality/ Rural Municipality is the lead agency of LEOC functioning in pre, during, and post- disaster situations.

1.2 Objective of Assessment

- To assess and generate information on the status of LEOCs of all the local levels of 32 earthquake affected districts,
- To identify needs and capacity gaps to establish and reinforce LEOC at local level for operationalizing the DRRM Act, 2017,
- To inform national policy making body MoHA/NDRRMA, MoFAGA about the latest status of LEOC, which will help them to informed decision making

1.3 Limitation of Assessment

One of the key limitations of LEOC Assessment is that data were collected remotely, i.e., via phone interviews. Secondly, only key municipal officials such as Chief Administrative Officer or Municipal DRR Focal Person were interviewed in the assessment indicating that the LEOC Assessment Report represents a broad range of perspectives of key municipal officials. Hence, some limitations in the assessment have resulted due to subjective nature of response and differences in understanding of questions among respondents of different technical expertise and knowledge on DRR. Moreover, this assessment has not tracked recent initiatives on DRR undertaken by the local governments.

This LEOC Assessment is conducted with few parameters of LEOC to generate baseline information on the status of LEOC in Nepal. Therefore, the LEOC Assessment Report will not provide detailed information on the overall functioning of Emergency Operation Centers (EOCs).

1.4 Structure of Report

The overall LEOC Assessment Report is structured into four chapters:

- The **first chapter** introduces the background of the study and indicates objectives of the assessment. It also provides key limitations of the study and outlines the structure of the report.
- The **second chapter** unfolds and elaborates on the tools and methodology used for the end-line.
- The **third chapter** presents quantitative analysis of assessment through tables and figures and findings derived from LEOC Assessment.
- The **four chapter** briefly summarizes conclusions from the LEOC Assessment and provides key recommendations.

Chapter II: Methodology

2.1 Desk Review & Study Design of LECO Assessment

Different available documents and guidelines on DRR were reviewed for designing questionnaires of assessment. The questionnaire primarily considered key points outlined in the DRRM Act 2017 that envisioned LEOC as a mandatory task for smooth execution of DRRM provision. Similarly, four priority actions of SFDRR were also considered in the scheming assessment form.

Cross-sectional study design approach was used for assessment with the information collected from the representatives and the officials from local governments that represented the initial status of LEOC from the cross-section of targeted local level. A close-ended questionnaire was developed and was composed of different section as presented in table below:

Table 1: Composition of LEOC Assessment Questionnaire

Area of Questionnaire	Description
Introduction	Introduction with respondents and taking consent
General Background	Information on respondents' name, organization, designation in the organization, and contact details
Information on DRR	Information on disaster of the Rural Municipality/Municipality over the past five years, LDCRP, LEPRP, EWS installed at local level, provision of damage need assessment,
Information of Municipality having LEOC	Information on LEOC establishment date, contact details of LEOC/DRR focal person, SOP made for LEOC, reporting frequency pre and during disaster, EWS linked with LEOC, simulation exercise conducted for LEOC strengthening, LEOC linked with DEOC/PEOC/NEOC and challenges faced for functioning of LEOC
Information of Municipality not having LEOC	Information on underlying causes that is hindering local government to establish LEOC, any planning to establish LEOC, and budget allocated for LEOC

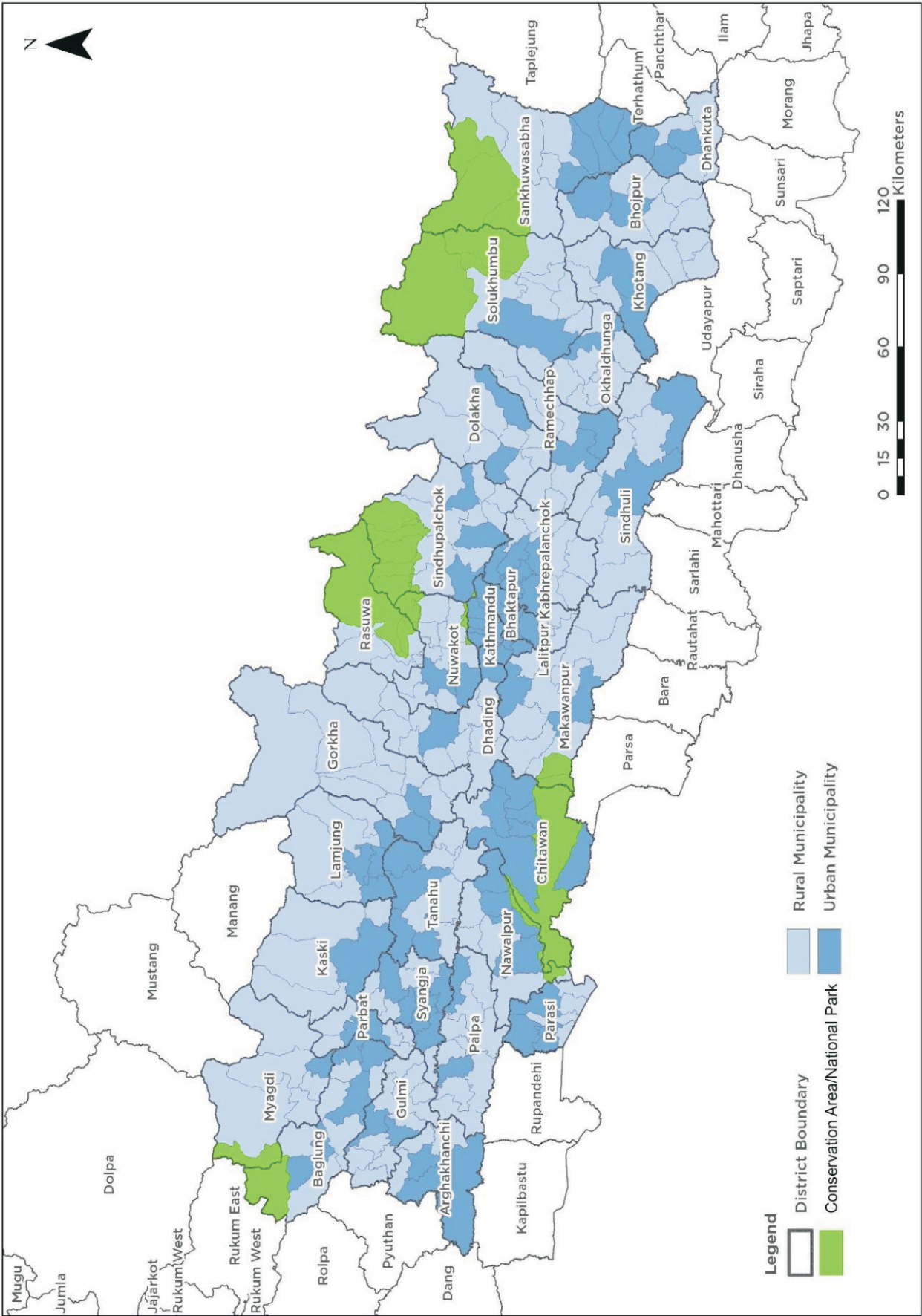
Source: LEOC Assessment, January 2021

2.2 Rural Municipality/Municipality Selection Criteria & Selection of Respondents

All the local governments i.e., 282 rural municipalities/municipalities of 32 earthquake affected districts were selected deliberately for the study because HRRP has presence and coverage within 32 EQ affected districts. HRRP has been doing regular coordination throughout 282 local governments of these districts. Overall, the assessment is successful in covering all types of local governments in Nepal, i.e., Metropolitan City, Sub-Metropolitan City, Municipality and Rural Municipality (Refer map 1, for municipality/rural municipality selected for study). Moreover, local governments selected for assessment belong from diverse geographical regions. Therefore, findings of assessment will be able to describe the position of DRR including LEOC's status and functionality at different local governments with reference to core versus periphery versus semi-periphery areas of Nepal. For instance, metropolitan city is considered as core, surrounding sub-metropolitan city and municipalities as periphery and rural municipalities as semi-periphery in the study to analyze the status of DRR at local level based on the access, remoteness, and resources this vast coverage also illustrates Rural/Urban scenario of the LEOCs, their status, needs, and gaps. In the assessment, it was agreed to interview any respondents representing a municipality/rural municipality including Chief Administrative Officer, DRR focal person, or municipal chair.

2.3 Data Collection & Analysis

Considering the pandemic situation due to COVID-19, telephone interviews were made to get required data for assessment. The LEOC Assessment was conducted beginning the first week of January 2021 until the second week of February 2021. Data gathered from assessment form were transferred to excel for cleaning and analysis. After that, a draft summary of findings was prepared and shared with the survey team to confirm the findings.



Chapter III: Analysis & Findings

This chapter is divided into three parts: Part 3.1 of the report provides general information of local governments. Part 3.2 highlights general disaster risk reduction. Part 3.3 provides the status of Local Emergency Operation Centers.

3.1 General Information of Local Governments

3.1.1 LEOCs Assessment Areas

LEOC Assessment was conducted in all the 282 local governments of 32 earthquake districts, where there are 186 rural municipalities and 96 urban municipalities which includes four metropolitan cities and one sub-metropolitan city (Figure 2). Based on this information, the existence, non-existence, and functional status of LEOC is analyzed in the following section 3.2 and 3.4.

Number of Local Governments Surveyed, (n)=282

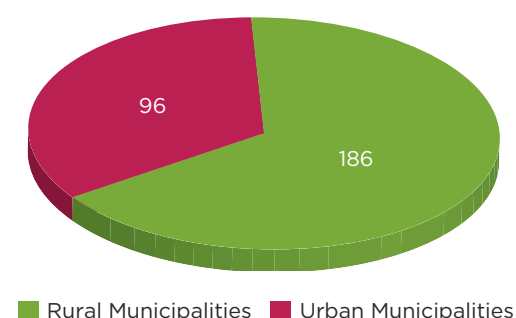


Figure 2: Number of Municipality Surveyed for LEOC Assessment

In 2017, on the recommendation of the Local Level Restructuring Commission, 753 local government units were established after the federal restructuring of local governments in 2017. There are 460 rural municipalities, 276 urban municipalities, 11 sub-metropolitan cities, and six metropolitan cities. As these local governments have recently formed, many of them are still working on strengthening their organizational capacity, physical infrastructure, and legal documents, policies, plan and guidelines.

3.1.2 Respondent's Designation

The main respondent of this assessment was the DRR focal person of local level. From this assessment, it is found that DRR focal persons of the local level include both government officials and elected local bodies. Elected local bodies are the political representatives elected by the people to represent them in a legislature, whereas a government official is a bureaucrat hired by the government to work for the public sector. Focal persons on DRR are designated from both types of officials in local governments, though the percentage of government officials is still high (84%), while the other 16% are DRR focal persons from the local elected body (16%) (Refer figure 3). These 84% of government

Respondent's Designation (n) = 282

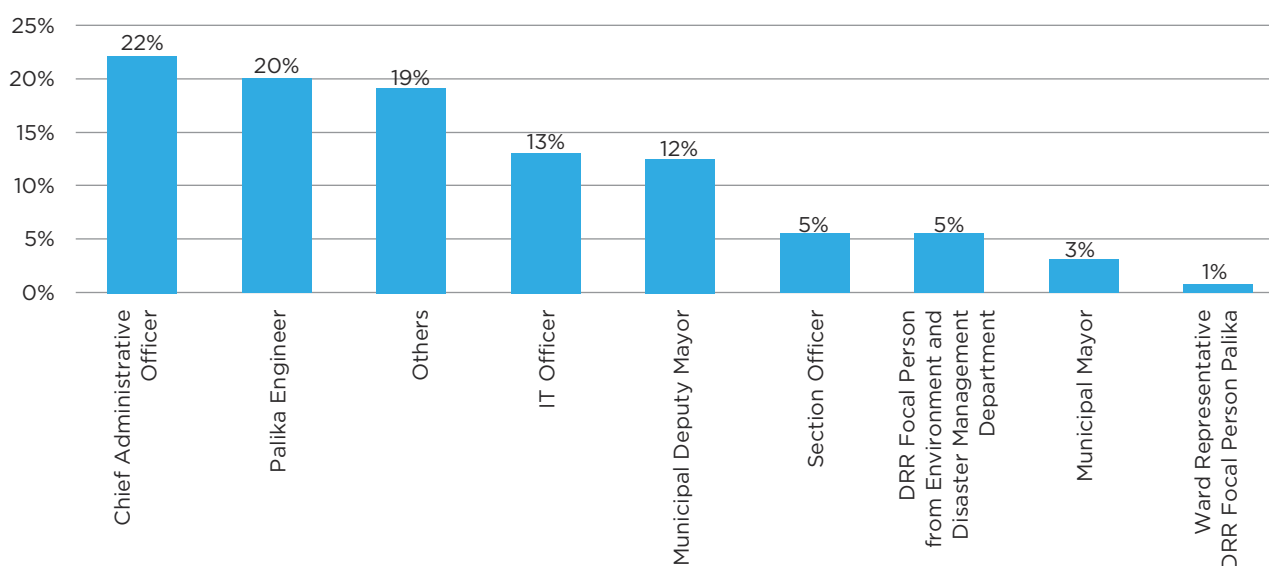


Figure 3: Designation of DRR focal persons at Local Levels

officials include Chief Administrative Officers (22%), Palika Engineers (20%), Others¹ (19%), IT Officers (13%), and Section Officers (5%). Likewise, 16% of respondents from local elected bodies include Municipal Chairperson (3%), Deputy Municipal Chairperson (12%), and Ward Chairperson (1%) (Refer figure 2). Hence, DRR focal person should be from government officials because all the administrative and development activities are being implemented by government officials and they are also leading the Environment and DRR Section at local levels.

3.2 Disaster Risk Reduction

3.2.1 Major Hazards at Local Level

Impact of disaster varies from Palika to Palika and within the community of same Palika because of diverse geographical structures, socio-cultural context, economic capacity, and communities' capacity to absorb shock and stresses of disaster. Considering this, respondents were asked about major local hazards. LEOC Assessment data reflects that four major hazards found in the surveyed Palika are, landslides (89%), flood (64%), earthquake (61%), and fire (50%). Subsequently, 30%

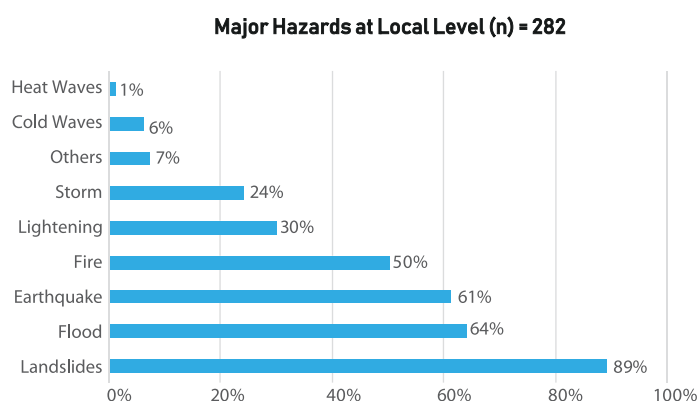


Figure 4: Major Hazards at Local Level

of the local governments further considered lightning and 24 % of them considered storms also as major hazards. Similarly, other hazards² and cold waves were also noted by 7% and 6% of the respondents subsequently (Figure 4). Further analysis of the data also demonstrates that landslides, floods, earthquakes, and fire are common hazards in both urban and rural municipalities (Refer annex 1 for more details). Though the hazards are local, survey data of four major local hazards are also aligned with disaster data presented by [MoHA published in June 2019](#). Disaster incident records of MoHA show that incidents of 3,973 fire hazards were recorded, followed by records of 482 landslides and 432 lightning hazards, and 418 flood hazards. (Refer to annex 1 to get Palika wise Hazards)

3.2.2 Disaster Vulnerability Rating of Local Governments

Based on the hazard incidents, local governments were asked to rate the vulnerability of their local level. Assessment data indicated that 69 respondents from local governments rated their Palika as highly vulnerable to disaster, 185 rated their Palika as medium, while 28 respondents considered their Palika as low to disaster risk (Figure 5). This rating is based on the perception and few indicators like frequency and intensity of hazard. Findings clearly indicate that vulnerability ratings of Palika need to be developed based on the scientific and systematic way by using common tools among different Palika. Each Palika level must have a DRM index. So that, with reference to DRM index, vulnerability and capacity status of different Palika can be compared as well as progress disaster risk reduction and management within the Palika can also be tracked periodically. (Refer to annex 1 to get Palika wise vulnerability rating)

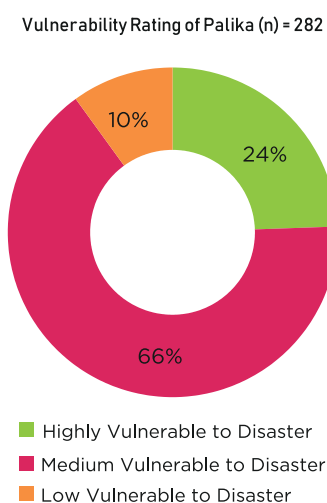


Figure 5: Vulnerability at Local Level

¹ Others are Planning Officer, Health Officer, & Accountant

² Others are COVID-19, Road Accidents, Soil Erosion, Wild Animal Attack, Snow fall

3.2.3 Availability of Planning for Disaster Risk Reduction & Emergency Response at Local Level

DRRM Act 2017 and Local Government Operations (LGOA) Act 2017 clearly stated roles and responsibilities of local government on developing local level planning, policy and guideline to reduce and manage disaster at local level and develop disaster and climate resilient communities. Referring to DRRM Act 2017 and LGOA 2017, local governments were further asked whether they have developed and endorsed LDCRP and LEPRP. LEOC Assessment data points out that 103 local governments have developed LDCRP, which include 56 rural municipalities and 47 urban municipalities (Figure 6).

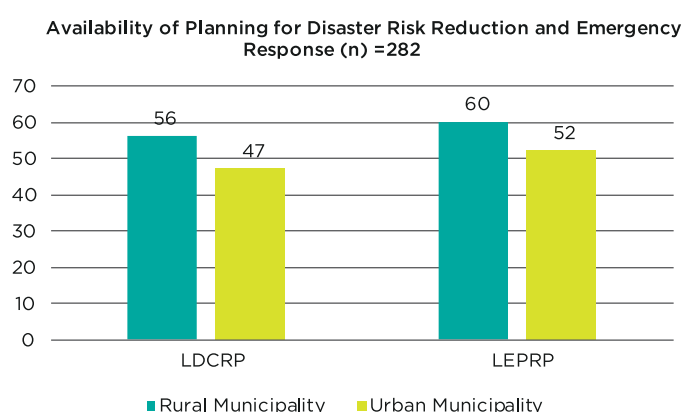


Figure 6: Availability of Planning for Disaster Risk Reduction and Emergency Response

Still, 63% (i.e., 179 out of 282) of the Palika have not started developing LDCRP. Since LDCRP addresses risk and vulnerability to disaster risks and climate change and incorporates key actions to be undertaken in all stages of disaster risk management – preparedness, prevention, mitigation, response, rehabilitation, reconstruction, recovery, and climate change adaptation, Local Governments need to take concerted action for preparing LDCRP for enhancing resilience to slow and rapid onset disaster. MoFAGA also encourages the local levels to prepare and endorse LDCRP to help local level identify disaster risk and vulnerability associated with disaster that the Palika are encountering. LDCRP will help the local level to prioritize disaster risk reduction activities, so that it can further support in contextualization of DRRM Act, 2017 at local level and help in mainstreaming DRR into development policies and planning processes.

Likewise, the LEPRP is a localized version of District Emergency Preparedness and Response Planning. The LEPRP is primarily focused on emergency response where responsibilities of different clusters are clearly defined with coordination mechanism, sectoral plan, and implementation. As per the LEOC data, 112 local governments including 60 rural municipalities and 52 urban municipalities have developed LEPRP (Figure 6).

Data indicates that 60% (i.e., 170 out of 282) of the local governments have not initiated developing LEPRP. As stated by the Chief Administrative Officer of Jwalamukhi Rural Municipality of Dhading, *“We don’t have a LEPR because of lack of knowledge and expertise to develop it. But we know that emergency preparedness and response plans are required for effective and prompt response can be carried out in a coordinated way so that scattered resources can be utilized to minimize the casualty and property loss. Our Palika has developed a relief distribution standard, which is the traditional practice exercised by Palika for immediate relief distribution during disaster.”*

Box 1: Opinion on DRR by Local Governments

"Our Palika is highly vulnerable to geological hazards i.e. landslides. Every year in monsoon, landslides disrupt human settlements and cause economic loss. Palika only has relief distribution guidelines for immediate response. Our Palika has realized the importance of DRR planning and emergency planning, but we are not able to develop it because of limited technical expertise."

-Chief Administrative Officer, Galchi Rural Municipality, Dhading

"Though our Palika has allocated a budget for DRR, they are not properly utilized and the allocated budget on DRR is about to freeze; Because our Palika lacks technical knowledge to implement DRR Activity."

-LDMC Focal Person, Bethanchowk Rural Municipality, Kavrepalanchowk

3.2.4 Provision of Damage Need Assessment in Post Disaster at Local Level

Effective emergency response depends on the capacity of local governments to quickly collect data, analyze, and determine immediate needs in post disaster scenarios. Considering this fact, local governments were questioned about provision of Damage Need Assessment (DNA) in their Palika, where 53 rural municipalities and 47 urban municipalities, i.e., in total 100 local governments stated having provision of DNA (Figure 7).

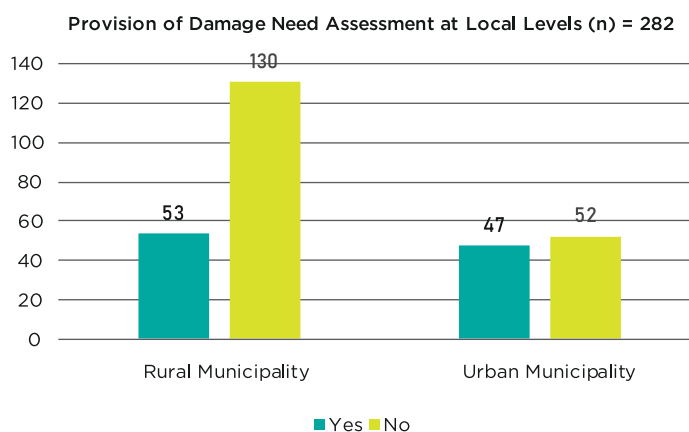


Figure 7: Provision of Damage Need Assessment at Local Levels

Undoubtedly, more than 64% (i.e., 182 of 282) of the local governments have not given priority to DNA. The 32

earthquakes affected districts are prone to multi-hazards and frequency and intensity of geological and meteorological hazards such as landslide, and flood/inundation etc. are during pre-monsoon and monsoon have been the major cause of human causality and economic loss. It is crucial that municipalities should have provision for DNA/Initial Rapid Assessment (IRA), and they also need to build capacity of staff, Disaster Management Committee, and local stakeholders in conducting DNA & IRA Depending on the intensity of the disaster, local governments should have a clear mechanism of linkages and coordination with the province and federal government to conduct Multi-Cluster Initial Rapid Assessment (MIRA) and PDNA.

3.2.5 Establishment of Early Warning System

Though every Palika has radio and television including newspapers, online newspapers, and social media for getting access to DRR related information, only 13 of the interviewed Palikas have established an Early Warning System (EWS), primarily focusing on flood and landslide hazards only (Figure 8). Eight local levels have developed Flood Early Warning System, one local level developed Landslide Early Warning System and three local levels stated having traffic signal as early warning system (Refer annex 3 to Details of Hazard Specific Early Warning System installed at Local Levels).

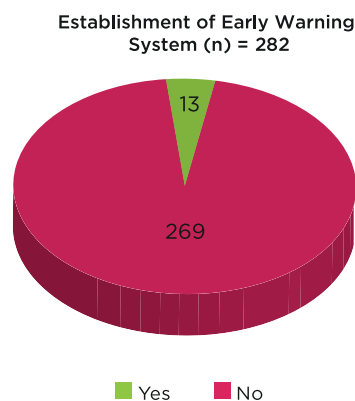


Figure 8: Establishment of EWS at Local Levels

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EWS is key for timely response during disaster and saving lives of people and reducing property damage from disaster. They required coordination efforts by Local Governments to establish a Multi-Hazards Early Warning System with the Department of Hydrology and Meteorology (DHM). Palika should explore possible weather-based stations in their areas together with DHM and information dissemination mechanisms should be developed and practiced, so that community people can access timely information and response on time.

Box 2: Opinion on Early Warning System by Local Governments

"If Tsho Rolpa Glacier Lake outburst, thousands of people residing downstream of Tamakoshi River will be adversely affected. Therefore, the Early Warning System at Tsho Rolpa Lake can save the lives of many people by evacuating them timely to safe locations after we receive early warning messages. "

-LDMC Focal Person, Manthali Municipality, Ramechhap

3.2.6 Activation of LEOC & Its Functionality

Indeed, this is a dismaying fact that LEOC is found in only 16 Local Governments in the 32 Earthquake affected districts. In both urban and rural municipalities, there are eight LEOCs established. Majority of the respondents mentioned that the reasons behind for not establishing LEOC in their Palika are due to limitations in technical knowledge, budget implication, and operational modality.

Table 2: LEOC & It's Functionality

LEOC Functionality	Establishment of LEOC (n=282)		LEOC Functional Palika (n=16)		LEOC Operational 24x7 hours/days in Palika (n=16)	
	Yes	No	Yes	No	Yes	No
Rural Municipality	8	175	7	1	6	2
Urban Municipality	8	91	7	1	3	5
Total	16	266	14	2	9	7

As per mandate of DRRM Act 2017, once the LEOC is active, it should be operated 24 hours X 7 day in shifts of 8- or 9-hours. However, if we look at the functionality of LEOC, only nine Local Governments are operational 24 hours X 7 days, though LEOC is fully functional in 14 local governments. Absence of LEOC in 266 local levels reflects that DRRM is not their major priority. Local governments' understanding and awareness on DRRM need to be enhanced along with capacity development to establish and operate LEOC. (Refer to Annex 2, to get Palika wise detail of LEOC)

3.2.7 Availability of Space and Equipment for LEOC

In accordance with the provision of DRRM Act 2017 and LGOA 2017, Local Governments are assigned to manage and arrange required location, working space, human and technical resources including essential services and facilities, equipment, and budget for establishing LEOC. Findings of this LEOC Assessment showed that 11 Local Governments (Refer Table 3) have provided separate offices for operating LEOC in which only nine local governments are fully equipped with IT equipment, furniture, and lifesaving equipment, whereas one local government is not equipped with lifesaving equipment. Without that equipment, LEOC cannot be fully functional and efficient. While designing and planning for LEOC, local governments need to allocate sufficient resources.

Table 3: Separate Office Space & Equipment Availability at LEOC

Municipality Types	Office Space Availability for LEOC (n) = 16		Equipment Availability (n)=16		Equipment Availability for LEOC (n) =10		
	Yes	No	Yes	No	IT Equipment	Lifesaving Equipment	Furniture
Rural Municipality	5	3	6	2	6	5	6
Urban Municipality	6	2	4	4	4	4	4
Total	11	5	10	6	10	9	10

3.2.8 Standard Operating Procedures for LEOC

Role of LEOC is significant for managing disaster risk and functioning of DRM structure at local level. Therefore, standard operating procedures (SOPs) are required for smooth functioning of LEOC. Assessment data illustrates that standard operating procedures have been formulated by LEOC of 11 Palikas only. Though 11 LEOC have formulated standard operating procedure, the majority of the Local Governments did not mention provision of reporting frequency before and during disaster. Respondents of six Palikas mentioned about having provision of reporting of hazard incidents including damage and loss caused due to disaster (Refer table 4). During a disaster, they normally do report to LDMC on an hourly, daily, and monthly basis, while in the normal time, they do report either monthly or on the need basis. These findings indicate that local governments are not aware of the SOP guideline for LEOC, which is designed and circulated and developed by MOHA. SOP helps in determining organizational procedures necessary to activate and operate EOCs quickly and efficiently. It also identifies the minimum resources requirements, and roles and responsibilities of the concerned officials for smooth operation of LEOC.

Table 4 Provision of SOP for LEOC & Provision of Reporting frequency in SOP

Municipality	Provision of SOP for LEOC (n=16)		Provision of reporting frequency in SOP (n=11)	
	Yes	No	Yes	No
Rural Municipality	5	3	3	2
Urban Municipality	6	2	3	3
Total	11	5	6	5

3.2.9 Capacity Building of LEOC

To test the functioning capacity of LEOC, simulation exercise plays an integral role for strengthening effective risk communication and improving response capacity. Out of 11 Palikas who have developed SOP, but only nine Palikas have been practicing simulation exercises either on an annual or semi-annual basis and two Palikas do not have provision for simulation exercise (Refer table 5). Two Palikas are doing simulation exercises on semi-annual basis, six Palika are doing annual basis, and one Palika is on need basis i.e., quoted in others section. This indicates still there is area to be improved in those Palikas who have established LEOC and developed SOP.

Finding also states that local governments have given less priority for capacity building on DRR. As a result, non or partial functioning of LEOC resulted in this study. For strengthening the capacity of local governments on LEOC, following activities like mock drill, simulation exercises will maximize institutional capacity to address and deal with managing and reducing disasters. In addition to this, simulation exercise and mock drill will also strengthen incident handling capacity and make LEOC updated and operational throughout the year.

Table 5: Capacity Building of LEOC

Municipality	Provision of Simulation Exercise for LEOC (n) =16		Period of simulation exercise (n) = 9			Future Planning for Simulation Exercise (n)=7	
	Yes	No	Semi-annually	Annually	Others	Yes	No
Rural Municipality	4	4	1	3	0	2	2
Urban Municipality	5	3	1	3	1	3	0
Total	9	7	2	6	1	5	2

3.2.10 Linkage of LEOC with DEOC, PEOC, and NEOC

Findings of the assessment reveal that LEOCs of only 10 Palikas have established linkage with EOCs at district, province, and national level (Figure 9). It is important to have linkage among all EOC for real time data collection, dissemination, and taking prompt action at all levels in an emergency situation, so that response can be on time, relief can be distributed without any delay, and lives of people and property can be saved.

Linkage of LEOC with DEOC, PEOC, and NEOC (n) = 16

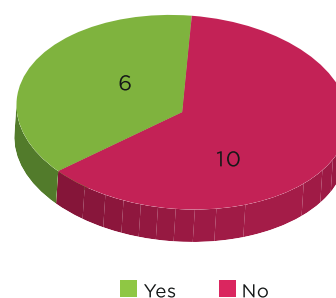


Figure 9: Linkage of LEOC with DEOC, PEOC and NEOC

Linkages of LEOC amongst different administrative and governing structures defined by Constitution of Nepal (Federal, Provincial, District, and Local) is not clearly mentioned in DRRM Act 2017 and policies. But considering the government structure and geological structure of Nepal, this study found LEOC is also equally important as NEOC and DEOC. On the other hand, resource constraints at local level were also clearly identified from this study, because of which 266 numbers of local levels still have not been initiated to establish LEOC. In this context, further discussion and advocacy with MoHA, MoFAGA, and MoCIT is still important to establish and operate LEOC and for linkage of LEOC with DEOC, PEOC, and NEOC.

3.2.11 Linkage of Early Warning System with LEOC

In Palika, where LEOC is functional such as Bigu Rural Municipality of Dolakha and Bharatpur Metropolitan City of Chitwan, they have established Flood Early Warning System and Dharche Municipality of Gorkha has established Landslide Early Warning System. Unfortunately, none of the LEOC established Palika have created linkage between EWS and LEOC. There should be linkage between EWS and LEOC for support to decision making at local, provincial, and federal Level and to disseminate early warning information up to the community level. In recent scenarios Flood Early Warning System has been established in major rivers of Nepal. DHM has also established a Forecast Based Early Warning System and disseminates such information through telecommunication systems, television, and radio including social media like Facebook. If such real time information is linked with LEOC, this will be supportive to disseminate information to the community people on time, so that the life of maximum number of people can be saved and also property loss of such vulnerable people can be minimized.

3.2.12 Challenges to Operate LEOC

Since new local governments under the federal system were recently elected and established in 2017, they are still struggling to strengthen their institutional capacity and the concept of disaster risk management is new to them. In the case of LEOC established Palikas, Local Governments are still having operational and technical challenges to function their LEOC smoothly (Figure 10). The operational challenges include limited human resources to

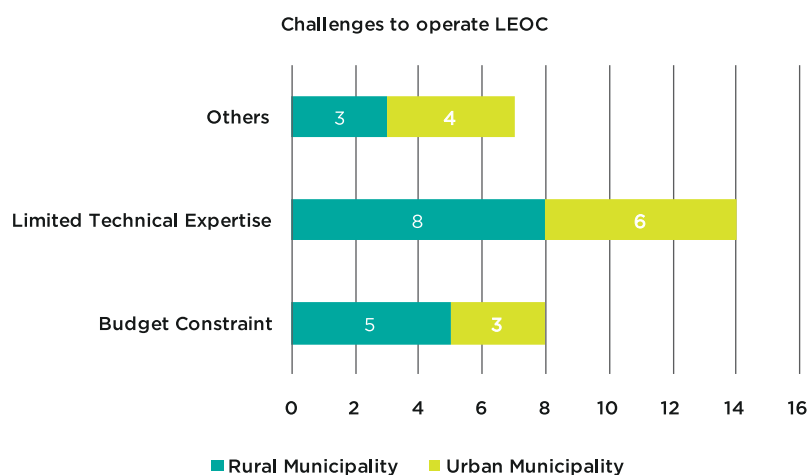


Figure 10: Challenges to Operate LEOC

operate 24x7 hours-days, lack of policies, guidelines and SOP at local level, unclear incident reporting and dissemination mechanism, and lack of awareness in the community about LEOC. Similarly, the technical challenges that local governments are facing include difficulties in getting bandwidth for VHF equipment from the Ministry of Communication and Information Technology (MoCIT), technical operational challenges, and limited technical human resources.

Box 3: Statement of LEOC Focal Person on challenges faced by Local Government for functionalization of LEOC

"The Ministry of Communication and Information Technology has not approved bandwidth for the operation of radio equipment."

-LEOC Focal Person, Shankharapur Municipality, Kathmandu

3.2.13 Prospects for LEOC at Local Level

The 266 local levels, that have not yet established LEOCs, were further asked whether they have any future planning for establishing LEOCs at local level. Data indicates that only 65 local levels have taken consideration for establishing LEOC, out of which only 19 local levels have made some planning with an allocation of budget for establishment of LEOC.

Table 6: Future Planning and Budget Allocation for LEOC Establishment

Prospects for LEOC	Planning for LEOC Establishment (n)=266		Budget Allocation for LEOC Establishment (n) = 65	
	Yes	No	Yes	No
Rural Municipality	35	140	10	25
Urban Municipality	30	61	9	21
Total	65	201	19	66

Still, 75% of the local level do not have any priority for establishing LEOCs. This finding indicates either they do not understand the importance of LEOCs for emergency response, rehabilitation, reconstruction, and recovery or they don't want to establish LEOCs. This shows, Local level needs capacity building on DRR intervention including awareness on the importance of LEOC. There also needs advocacy for DRR resilient Palika through minimization of risk of human causality, property losses from future disaster in respective Palikas.

As per SOP shared by Nilkantha Municipality, LEOC is being regulated and managed by the Local Disaster Management Committee (LDMC). LDMC is an authorized committee for disaster risk reduction and management and response at local level. Hence, regulation and management of LEOC under LDMC may be a good example for those Palikas, who are planning to establish LEOC.

Chapter IV: Conclusion & Recommendation

This chapter is divided into two sections. The first section 4.1 gives the conclusion of the report and in the subsequent section 4.2, recommendation is provided in the context of SFDRR.

4.1 Conclusion

Based on analysis of the previous chapter, it can be concluded that though DRRM Act and LOGA 2017 have clearly defined roles and responsibilities of local governments for managing disaster risk of local level, they are still having challenges to comply with their roles and responsibilities on DRR. Local governments also lack technical expertise on DRR for contextualization of acts, policies, & guidelines circulated by the federal government as per the local needs. Conclusion drawn from this assessment, is that local governments are facing challenges to take the lead on disaster risk management initiatives including operationalization of LEOCs. This is clearly signified through absence of LEOCs at many local levels in earthquake affected districts. Even though LEOCs are established in some of the rural municipalities and urban municipalities, local governments have had operational and technical challenges for its proper functioning. Inadequate funds allocation, limited human resources, and lack of technical knowledge are the major causes for the absence of LEOCs. Role of LEOC is crucial in making risk informed and prompt decisions in time of disaster by the federal, province, and local governments. This has not been clearly understood by the local level. Local Governments' leadership on DRR is substantial for risk governance and promoting resilience of local level. Therefore, local governments need capacity building support on strengthening DRR including LEOC establishment, so that they can manage their local hazards, reduce the impact of disasters, and lead towards disaster resilient communities.

4.2 Recommendation

In this section, recommendations for strengthening LEOCs are provided based on the analysis and findings of the assessment and priority areas of the SFDRR. In this regard, recommendations for strengthening operationalization of LEOC are provided.

Recommendation for Local Governments

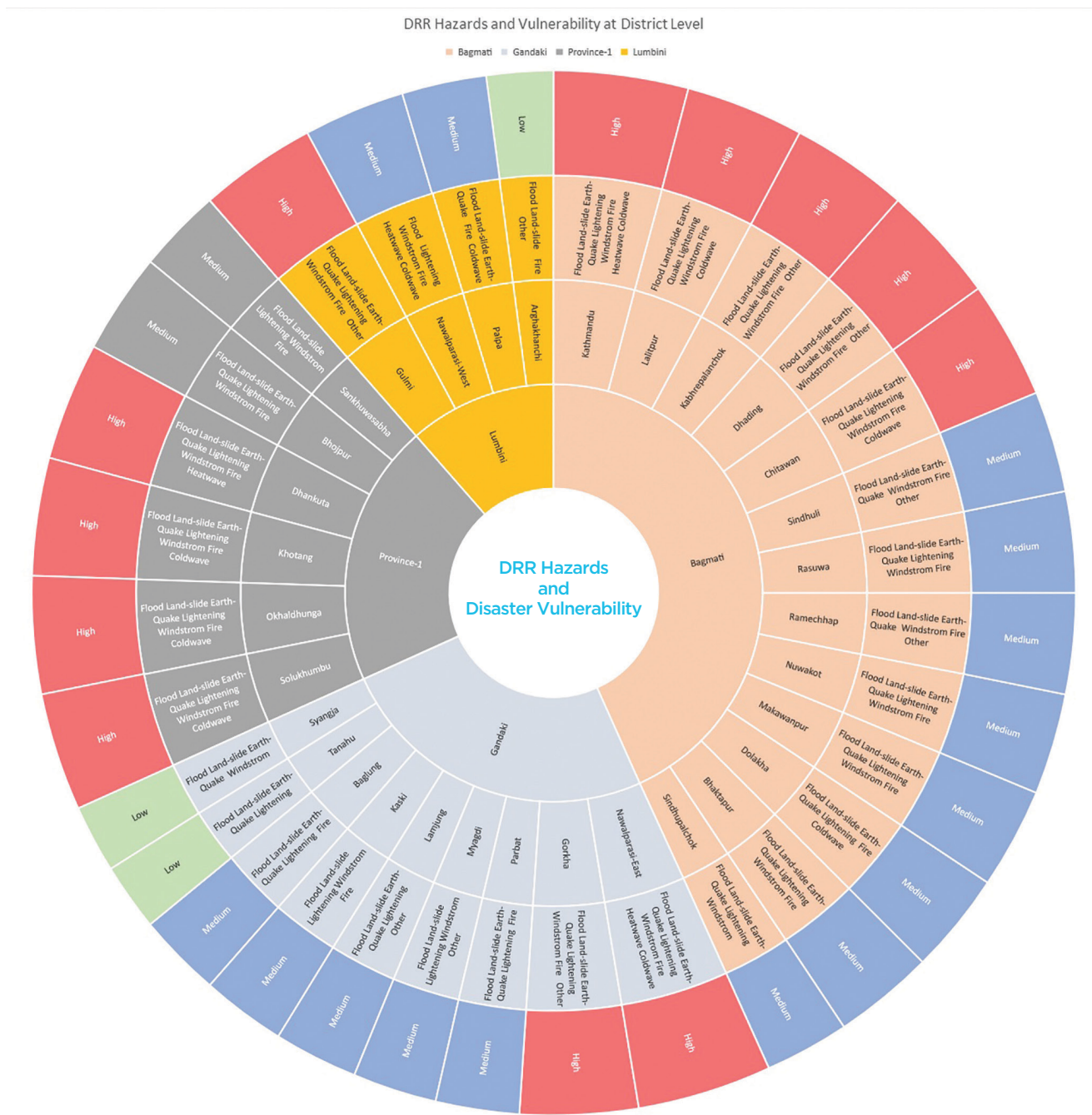
- Though LEOC is prioritized by the federal government, 83% of local governments have not initiated to establish LEOC in their respective Palikas. Therefore, establishment of LEOC should be a priority of local governments, and DRR practitioners working in respective local areas.
- Linkages between community, ward, LEOC, DEOC, PEOC and NEOC should be clearly established to capture the incident from the community and flow upwards through EOC in real time for effective and efficient response.
- Arrange Simulation Exercises on annual or semi-annual basis to evaluate the functionality of LEOCs, update the technical knowledge of DRR focal persons and to test the LEOC as a whole system to find the gaps and update the system.
- It is a proven statement that early warning systems (EWS) save the lives of people if EWS is linked with LEOC, disseminate warnings timely and responded to by the community on time.
- Role of the LEOCs in different phases like normal phase, alert phase, response phase and recovery phase should be clearly defined in SOPs and strengthen the capacity of LEOCs accordingly.
- Recommendation for establishment of linkages between LEOC with BIPAD portal
- Develop GIS data base with disaggregated information of Palika and link GIS data base with LEOC.
- As recommended by LEOC Focal Person of Changunarayan Municipality, LEOC can share the same office space with IT section Palika for regular operation of LEOC and its sustainability.

Recommendations for National Government (MoHA, NDRRMA)

- Render technical support to establish LEOC and support.
- Support for strengthening alternative communication systems and linkages among EOCs of Nepal,
- Coordination and technical support to award licenses for radio communication to the local governments.
- Developing and implementing the monitoring mechanism of LEOCs for its smooth operation.

Annexes

Annex 1: Palika Wise Hazards and Vulnerability Rating



Annex 2: Palika Wise Detail of LEOC & Its Functionality

District	Local Governments Name	LEOC Established Date (B.S)	LEOC Focal Person	Contact Number of Focal Person	LEOC Functionality	LEOC Operational 24/7 hours-days	Specific Duty Assigned for LEOC	Separate Office Space for Palika	Standard SOP developed for LEOC	Linkages of LEOC with DEOC/PEOC/NEOC	Availability of IT Equipment/Furnitures/Emergency Equipment at LEOC
Chitwan	Bharatpur Metropolitan City	2076	Bal Krishna Thapaliya	9845051127	Yes	Yes	Yes	Yes	No	Yes	Yes
Makwanpur	Bhimphedi Rural Municipality	2075	Shiva Raj Rumba	9855030630	Yes	Yes	Yes	No	No	Yes	Yes
Nuwakot	Bidur Municipality	2075	Subash Pandey	9841548916	Yes	Yes	Yes	Yes	Yes	No	Yes
Dolakha	Bigu Rural Municipality	2074	Yudhishthir Khadka		Yes	Yes	Yes	No	Yes	Yes	No
Bhaktapur	Changunarayan Municipality	2075	Siddhanta Neupane	9843723924	Yes	No	No	Yes	Yes	Yes	No
Gorkha	Dharche Rural Municipality	2076	Chief Administrative Officer	9851164646	Yes	No	No	No	Yes	No	Yes
Ramechhapp	Gokulganga Rural Municipality	2076	Parbat phuyal	9863023018	Yes	Yes	Yes	Yes	Yes	No	Yes
Makwanpur	Kailash Rural Municipality	2077	Shyam Sundar Mahato	9854037169	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kathmandu	Kathmandu Metropolitan City	2074	Badri Nagarkoti	9851037087	Yes	No	No	Yes	Yes	Yes	No
Lamjung	Kwholasothar Rural Municipality	2076	Chittra Ghale	9846074601	No	No	No	yes	Yes	No	No
Lalitpur	Lalitpur Metropolitan City	2076	Harish Chandra Lamichhne	9851052073	Yes	No	No	Yes	Yes	Yes	Yes
Kathmandu	Nagarjuna Municipality	2076	Rajendra Prasad Gautam	9851218076	No	No	No	No	Yes	No	No
Makwanpur	Raksirang Rural Municipality	2075	Govind Adhikari	9855076641	Yes	Yes	Yes	Yes	No	Yes	Yes
Chitwan	Ratnanagar Municipality	2073	Bishwas Sheetha	9845140124	Yes	Yes	Yes	Yes	No	Yes	Yes
Kathmandu	Shankharapur Municipality	2075	Prakash Singh Saud	9849898933	Yes	No	No	No	Yes	No	No
Bhojpur	Temkemaityung Rural Municipality	2076	Lila Prasad Parajuli	9842084600	Yes	Yes	Yes	Yes	No	Yes	Yes

Annex 3: Details of Hazards Specific Early Warning System Installed at Local Levels

District	Local Level	Types of EWS Installed
Dolakha	Bigu Rural Municipality	Flood Early Warning System
Gorkha	Dharche Rural Municipality	Landslide Early Warning System
Dolakha	Gaurishankar Rural Municipality	Flood Early Warning System
Dolakha	Baiteshwor Rural Municipality	Flood Early Warning System
Makawanpur	Hetauda Sub-Metropolitan City	Traffic Signal
Chitwan	Bharatpur Metropolitan City	Traffic Signal
Palpa	Tinau Rural Municipality	Flood Early Warning System
Gulmi	Musikot Municipality	Flood Early Warning System
Gulmi	Gulmidarbar Rural Municipality	Flood Early Warning System
Nawalparasi-East	Devchuli Municipality	Traffic Signal
Nawalparasi-East	Gaidakot Municipality	Flood Early Warning System
Nawalparasi-East	Kawasoti Municipality	Flood Early Warning System

Annex 4: LEOC Assessment Questionnaire

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MY NAME IS I AM FROM HRRP, AND I AM WORKING ASHRRP IS CONDUCTING INITIAL ASSESSMENT ON LEOC. LEOC A PHYSICAL FACILITY DESIGNATED FOR THE GATHERING AND DISSEMINATION OF INFORMATION, IN ADDITION TO CARRYING OUT CONTINUOUS DISASTER SITUATION ANALYSIS AT LOCAL LEVEL. I WOULD LIKE TO REQUEST YOU TO PROVIDE YOUR VALUABLE TIME TO PROVIDE INFORMATION ON LEOC ASSESSMENT.

Read out to the respondent

PLEASE TAKE THE CONCENT FROM THE RESPONDENT. *

Respondent's CONSENT

☐

Yes (छै)

☐

No (छैन)

Respondent Detail

PLEASE ENTER THE NAME OF RESPONDENT (उत्तरदाताको नाम) *

PLEASE ENTER THE POSITION OF RESPONDENT (उत्तरदाताको पद) *

PLEASE ENTER THE CONTACT NUMBER OF RESPONDENT (उत्तरदाताको सम्पर्क नम्बर) *

PLEASE ENTER THE PROVINCE NAME (प्रदेशको नाम) *

☐

Province 1

☐

Province 2

☐

Bagmati

☐

Gandaki

☐

Province 5

☐

Karnali

☐

Sudurpaschim

PLEASE ENTER THE DISTRICT NAME (जिल्लाको नाम) *

PLEASE ENTER THE MUNICIPALITY NAME (पालिकाको नाम) *

DRR Questionnaire (विपद् जोखिम न्युनिकरण सम्बन्धी प्रश्नावलीहरु)

WHAT ARE THE MAIN HAZARDS OF YOUR PALIKA IN THE PAST FIVE YEARS? (विगतको पाँच वर्षमा तपाईंको पालिकाको मुख्य प्रकोपहरु के के हुन ?) *

<https://enketo.ona.io/x/yb7dCVCQ>

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- ☐ Flood (बाढि)
- ☐ landslide (पहिरो)
- ☐ earthquake (भुकम्प)
- ☐ lightening (चट्यांग)
- ☐ windstrom (हावाहुरी)
- ☐ fire (आगलागी)
- ☐ heatwave (लु (तातो हावा))
- ☐ coldwave (शितलहर)
- ☐ Other (अन्य)

IF OTHERS, PLEASE MENTION OTHER.(अन्य यदि अन्य भए, कृपया उल्लेख गर्नुहोस्) *

HOW DO YOU RATE DISASTER OF YOUR PALIKA?(तपाईं आफ्नो पालिकाको विपद्लाई कुन तहमा राख्नुहुन्छ ?) *

- ☐ High (उच्च)
- ☐ Medium (मध्यम)
- ☐ Low (निम्न)

DOES YOUR PALIKA HAVE LOCAL DISASTER CLIMATE RESILIENCE PLANNING (LDCRP) OR LOCAL DISASTER RISK MANAGEMENT PLANNING?(तपाईंको पालिकामा स्थानीय विपद् जलवायु उत्थानशील योजना वा स्थानीय विपद् जोखिम न्युनिकरण योजना छ ?) *

- ☐ Yes (छ)
- ☐ No (छैन)

DOES YOUR PALIKA HAVE LOCAL EMERGENCY PREPAREDNESS AND RESPONSE PLANNING (LEPRP) OR MUNICIPAL EMERGENCY PREPAREDNESS AND RESPONSE PLANNING (MEPRP)?(तपाईंको पालिकामा स्थानीय विपद् आपत्कालिन पूर्वतयारी तथा प्रतिकार्य योजना वा पालिका विपद् आपत्कालिन पूर्वतयारी तथा प्रतिकार्य योजना छ ?) *

- ☐ Yes (छ)
- ☐ No (छैन)

DOES YOUR PALIKA HAVE GUIDELINES/INSTITUTIONAL ARRANGEMENT TO CONDUCT RAPID NEED ASSESSMENT POST DISASTER?(तपाईंको पालिकामा विपद् पश्चात् तत्काल गरिने दुर्त आवश्यकता मुल्यांकनको (RAPID NEED ASSESSMENT) लागि निदेशिका वा संस्थागत व्यवस्था छ ?) *

- ☐ Yes (छ)

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☐ Yes (छ) ☐ No (छैन)	
IS EARLY WARNING SYSTEM INSTALLED IN YOUR PALIKA ?(के तपाईंको पालिकामा पूर्व सूचना प्रणालीको व्यवस्था छ ?) *	
☐ Yes (छ) ☐ No (छैन)	
IF YES, EARLY WARNING SYSTEM IS INSTALLED FOR WHICH HAZARDS (यदि छ भने, पूर्व सूचना प्रणाली कुन प्रकोपको लागि जडान गरिएको हो ?) *	
☐ Flood ☐ Landslide ☐ Other	
IF OTHERS, PLEASE MENTION ((अन्य यदि अन्य भए, कृपया उल्लेख गर्नुहोस्) *	
.....	
IF NO, WHAT ARE THE REASONS? (यदि छैन भने, पूर्व सूचना प्रणाली जडान नगरिनुका कारणहरु के के हुन् ?)	
.....	
LEOC Questionnaire (स्थानीय आपत्कालिन् कार्यसंचालन् केन्द्र सम्बन्धी प्रश्नावलीहरु)	
HAS THIS PALIKA ESTABLISHED LOCAL EMERGENCY OPERATION CENTER (LEOC)? (के तपाईंको पालिकामा स्थानीय आपत्कालिन् कार्यसंचालन् केन्द्र छ ?) *	
☐ Yes (छ) ☐ No (छैन)	

<https://enketo.ona.io/x/yb7dCVCQ>

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LEOC Having Palika Questionnaire(पालिकामा स्थानीय आपत्कालिन् कार्यसंचालन केन्द्र व्यवस्था भएमा सोधिने प्रश्नावलीहरु)

WHEN WAS LEOC ESTABLISHED?(तपाईंको पालिकामा स्थानीय आपत्कालिन् कार्यसंचालन केन्द्र कहिले स्थापना भएको हो ?)

*

WHAT IS THE NAME OF LEOC OR DRR FOCAL PERSON?(LEOC OR DRR को लागि पालिकामा सम्बन्धित व्यक्तिको नाम के हो ?)

*

WHAT IS THE CONTACT NUMBER OF LEOC OR DRR FOCAL PERSON?(LEOC OR DRR को सम्पर्क नम्बर के हो ?)

WHAT IS THE EMAIL ADDRESS OF LEOC OR DRR FOCAL PERSON? (LEOC OR DRR को ईमेल ठेगाना के हो ?)

IS LEOC FUNCTIONAL ?(के पालिकामा रहेको स्थानीय आपत्कालिन् कार्यसंचालन केन्द्र संचालनमा छ ?)

*

- ☐ Yes (छ)
- ☐ No (छैन)

DOES IT OPERATE 24/7?(के स्थानीय आपत्कालिन् कार्यसंचालन केन्द्र सातै दिन/चौबिसै घण्टा संचालनमा रहन्छ ?)

*

- ☐ Yes (छ)
- ☐ No (छैन)

DOES YOUR PALIKA ASSIGN DUTY FOR 24/7 LEOC OPERATION? (के स्थानीय आपत्कालिन् कार्यसंचालन केन्द्र संचालन को सातै दिन चौबिसै घण्टाको लागि मानवीय स्रोतको व्यवस्थापन गरिएको छ ?)

*

- ☐ Yes (छ)
- ☐ No (छैन)

DOES PALIKA HAVE SPERATE OFFICE SPACE FOR LEOC (के स्थानीय आपत्कालिन् कार्यसंचालन केन्द्र संचालनको अलग कार्यलय कोठाको व्यवस्था गरिएको छ ?)?

*

- ☐ Yes (छ)

<https://enketo.ona.io/x/yb7dCVCQ>

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☐ No (छैन)	
DOES PALIKA HAVE STANDARD OPERATIONG PROCEDURE (SOP)/GUIDELINE OF LEOC?(के स्थानीय आपत्कालीन् कार्यसंचालन केन्द्र संचालनको कार्यसंचालन विधि (SOP) वा निदेशिका तयार पारिएको छ ?)	*
☐ Yes (छ) ☐ No (छैन)	
DOES SOP OF LEOC HAS PROVISION FOR REPORTING IN DISASTER?(के स्थानीय आपत्कालीन् कार्यसंचालन केन्द्र संचालनको कार्यसंचालन विधि (SOP) वा निदेशिकामा विपद्को समय गरिनुपर्ने प्रतिवेदन तालिका उल्लेख गरेको छ ?)	*
☐ Yes (छ) ☐ No (छैन)	
REPORTING FREQUENCY DURING DISASTER (विपद्को बेला गरिने प्रतिवेदन तालिका)	*
☐ Hourly (घण्टा) ☐ Daily (दैनिक) ☐ Weekly (साप्ताहिक) ☐ Monthly (मासिक) ☐ Quarterly (चौमासिक) ☐ Half Yearly (अर्धवार्षिक) ☐ Anually (वार्षिक) ☐ Other (अन्य)	
IF OTHERS, PLEASE MENTION OTHERS (अन्य यदि अन्य भए, कृपया उल्लेख गर्नुहोस्)	*
REPORTING FREQUENCY DURING NORMAL SITUATION (सामान्य अवस्थामा प्रतिवेदन तालिका)	*
☐ Hourly (घण्टा) ☐ Daily (दैनिक) ☐ Weekly (साप्ताहिक) ☐ Monthly (मासिक) ☐ Quarterly (चौमासिक) ☐ Half Yearly (अर्धवार्षिक) ☐ Anually (वार्षिक) ☐ Other (अन्य)	

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IF OTHERS, PLEASE MENTION OTHERS (अन्य यदि अन्य भए, कृपया उल्लेख गर्नुहोस्)		*
DID YOUR PALIKA ORGANIZE SIMULATION EXERCISE OF LEOC IN PAST? (के तपाईंको पालिकाले स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालनको नमुना अभ्यास गराएको छ ?) ☐ Yes (छ) ☐ No (छैन)		*
IF YES, IN WHICH PERIOD OF TIME? (यदि तपाईंको पालिकाले स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालनको लागि नमुना अभ्यास गराएको छ भने, कति समय अवधिमा नमुना अभ्यास गराइन्छ ?) ☐ 6 Months (६ महिना) ☐ 1 Year (१ वर्ष) ☐ Other (अन्य)		*
IF OTHERS, PLEASE MENTION OTHERS (अन्य यदि अन्य भए, कृपया उल्लेख गर्नुहोस्)		*
IF NO, DO YOU HAVE ANY PLAN TO ORGANIZE SIMULATION EXERCISE IN FUTURE?(यदि छैन पालिकाले स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालनको लागि नमुना अभ्यास गर्नको लागि कुनै योजना बनाएको छ?) ☐ Yes (छ) ☐ No (छैन)		*
DOES YOUR LEOC IS FULLY EQUIPPED? (के तपाईंको पालिकामा स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालन गर्नका लागि आवश्यक पर्ने सामग्रीहरु छन् ?) ☐ Yes (छ) ☐ No (छैन)		*
WHAT ARE THE EQUIPMENT AVAILABLE ?(आवश्यक सामग्रीहरु के के हुन् ?) ☐ Furniture (फर्निचरहरु) ☐ IT Equipment (IT उपकरणहरु) ☐ Life saving Equipment (जीवन रक्षाको लागि उपकरणहरु) ☐ Other (अन्य)		*

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IF OTHERS, PLEASE MENTION OTHERS (अन्य याद अन्य भए, कृपया उल्लेख गर्नुहोस्)

WHAT ARE THE FURNITURE AVAILABLE?(कस्ता फर्निचरहरु उपलब्ध छन् ?) *

Please mark all that are available

- ☐ Computer Table
- ☐ Working Table
- ☐ Office Chair
- ☐ Meeting chair
- ☐ Meeting table
- ☐ Radio Room Table
- ☐ Filing cabinet

WHAT ARE THE IT EQUIPMENT AVAILABLE? (कस्ता IT उपकरणहरु उपलब्ध छन् ?) *

Please mark all that are available

- ☐ Desktop Computer
- ☐ Laptop
- ☐ Multimedia Projector
- ☐ Digital Camera
- ☐ Webcam
- ☐ Headphone
- ☐ Router
- ☐ HF Radio Set
- ☐ GPS Device

WAT ARE THE LIFE SAVING EQUIPMENT AVAILABLE? *

Please mark all that are available

- ☐ lightning resistance (wiring & earthing)
- ☐ Safety helmet
- ☐ rubber boots
- ☐ rescue boots
- ☐ life jacket
- ☐ shovel
- ☐ pick axe
- ☐ crobar
- ☐ metal bucket
- ☐ waist bag

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- ☐ stretcher
- ☐ cap
- ☐ gloves
- ☐ Tarpaulin
- ☐ Blanket
- ☐ first aid kit
- ☐ water purifier liquid
- ☐ emergency light
- ☐ metal trunk
- ☐ portable concrete cutter
- ☐ bolt cutter
- ☐ portable fire fighting pump
- ☐ rescue hammer
- ☐ static rope
- ☐ fire rope ladder
- ☐ screw gate carabiner
- ☐ Flashlight/lamp for forehead
- ☐ Rechargeable batteries and extra lamps
- ☐ Water resistant torches
- ☐ Whistles
- ☐ Hand-held/portable loud speaker
- ☐ Scissors/Utility knives
- ☐ Hi-Vi Jackets

IS EARLY WARNIG SYSTEM LINKED WITH LEOC?(के पूर्व सूचना प्रणालीलाई स्थानीय आपत्कालीन् कार्यसंचालन केन्द्रमा समावेश गरिएको छ ?)

*

- ☐ Yes (छ)
- ☐ No (छैन)

IS LEOC LINKED WITH DEOC/PEOC/NEOC? (के पालिकामा रहेको स्थानीय आपत्कालीन् कार्यसंचालन केन्द्रको जिल्ला, प्रदेश र राष्ट्रिय आपत्कालीन् कार्यसंचालन केन्द्र सम्बन्ध छ?)

*

- ☐ Yes (छ)
- ☐ No (छैन)

WHAT ARE THE CHALLENGES YOU FACED FOR REGULARLY FUNCTIONING OF

*

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LEOC?तपाईंको पालिकामा स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालन गर्नमा देखा परेका समस्याहरु के के हुन् ??

- ☐ Budget Constraint (बजेटको अभाव)
- ☐ Lack of Technical expertise (प्राविधिक मानविय स्रोतको कमी)
- ☐ Other (अन्य)

IF OTHERS, PLEASE MENTION OTHERS (अन्य यदि अन्य भए, कृपया उल्लेख गर्नुहोस्)

*

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LEOC not having palika questionnaire (पालिकामा स्थानीय आपत्कालीन कार्यसंचालन केन्द्र व्यवस्था नभएमा सोधिने प्रश्नावलीहरु)

WHAT IS THE REASON BEHIND FOR NOT ESTABLISHING LEOC IN YOUR RM/MUNICIPALITY?(तपाईंको पालिकामा स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालन नआउनका कारणहरु के के हुन् ?)

- ☐ Budget Constraint (बजेटको अभाव)
- ☐ Lack of Technical expertise (प्राविधिक मानविय स्रोतको कमी)
- ☐ Other (अन्य)
- ☐ I am not aware of LEOC (LEOC जानकारीमा नभएको)

IF OTHERS, PLEASE MENTION OTHER(अन्य यदि अन्य भए, कृपया उल्लेख गर्नुहोस्)

DOES YOUR PALIKA HAVE ANY PLANNING TO OPERATE LEOC?(तपाईंको पालिकाले स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालन गर्नको लागि कुनै योजना बनाएको छ ?)

- ☐ Yes (छ)
- ☐ No (छैन)

IF YES, DO YOU HAVE ALLOCATED BUDGET FOR LEOC FORMATION?(यदि छ भने, पालिकाले स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालन गर्नको लागि बजेट बिनियोजन गरेको छ ?)

- ☐ Yes (छ)
- ☐ No (छैन)

IF YES, WHAT IS THE BUDGET ALLOCATED FOR LEOC FORMATION? (यदि पालिकाले स्थानीय आपत्कालीन कार्यसंचालन केन्द्र संचालन गर्नको लागि बजेट बिनियोजन गरेको छ भने बिनियोजित बजेट उल्लेख गर्नुहोस् ?)

Remarks

PLEASE FEEL FREE TO ADD THE THINGS THAT YOU FIND AND OBSERVE DURING THE SURVEY.

Enumerator Detail

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PLEASE ENTER THE NAME OF ENUMERATOR (सर्वे गर्ने व्यक्तिको नाम)	*
PLEASE ENTER THE ORGANIZATION NAME OF ENUMERATOR (सर्वे गर्ने व्यक्तिको संस्था)	*
PLEASE ENTER THE DESIGNATION OF ENUMERATOR (सर्वे गर्ने व्यक्तिको पद)	*
PLEASE ENTER THE CONTACT NUMBER OF ENUMERATOR (सर्वे गर्ने व्यक्तिको सम्पर्क नम्बर)	*

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Annex 5: “Stepping Towards DRR Resilient Municipality-A Case Story of Changunarayan Municipality”

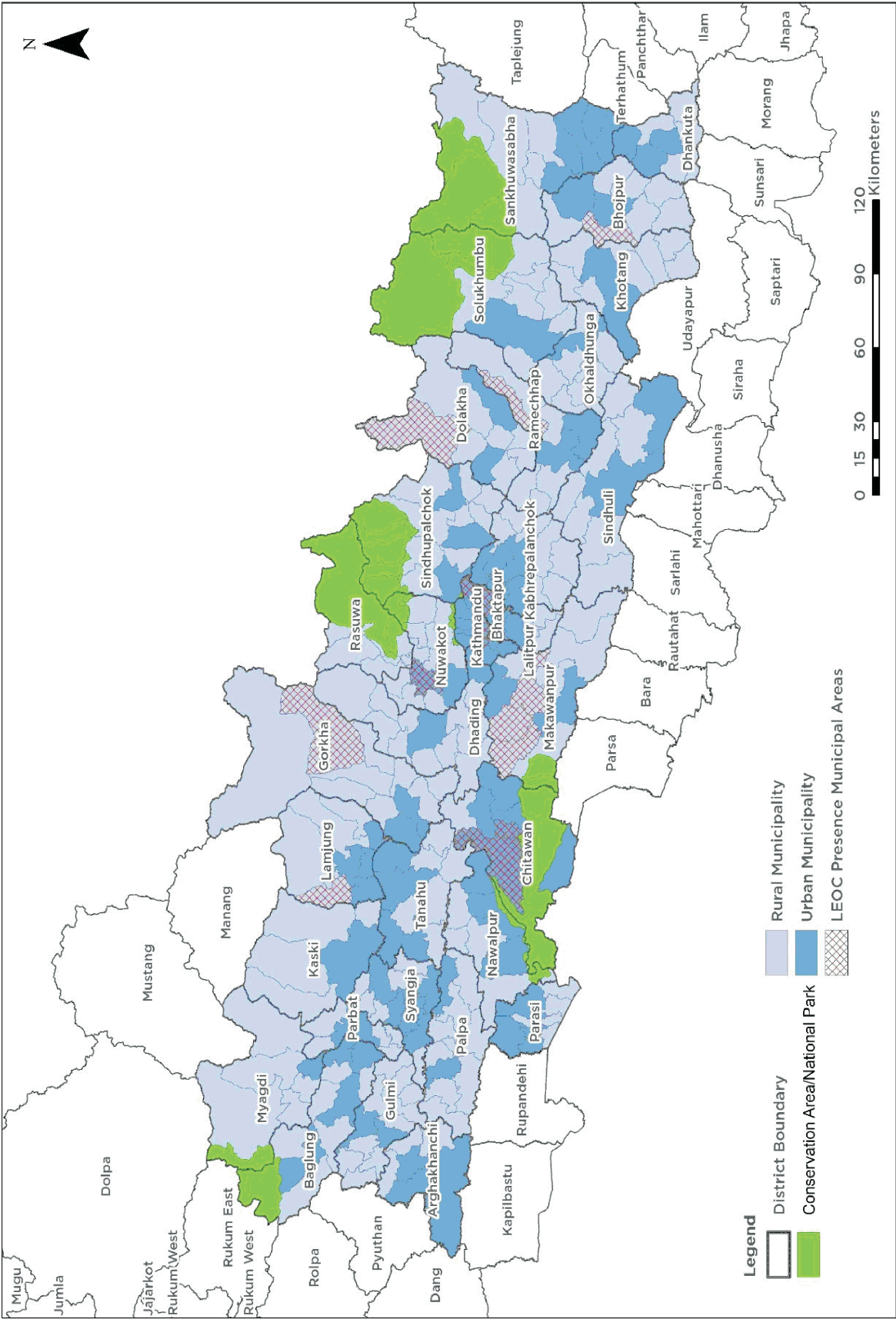
Historically significant and naturally beautiful, Changunarayan Municipality is one of the largest municipalities of Bhaktapur district. The municipality is named after the oldest historic temple of Nepal, “Changu Narayan Temple”, built in Lichhavi era. Changunarayan Municipality was one of the worst affected municipalities of Bhaktapur district by the 2015 earthquake. Besides earthquakes, landslides, fire, and lightning are the major hazards of this municipality. Every year locals have to encounter trauma from those hazards and there was no escaping from it. However, gradually, the scenario of the municipality has changed, and the municipality is delivering quality services on DRR.

The Municipality has been implementing numerous disaster preparedness and mitigation work to strengthen DRR capacity of Palika. In the fiscal year 2077/78 B.S., Municipality has allocated NPR 1 crore fund for disaster mitigation work like landslide protection wall construction and bio-engineering wall construction in the flood prone areas. Additional NPR 1 crore budget is also allocated for the DRR fund. DRR volunteers of the community are also provided with capacity building opportunities on fire hazard safety and flood hazard preparedness. The municipality is also doing regular updates of LDCRP and LEPRP after doing VCA exercises. Current work priorities of the municipality are to functionalize Municipal EOC (MEOC) 24x7 hours/days and they are planning to make it functional with the support of the City Police. Engineering design of the safe evacuation center is almost completed, and they are working on the construction of a safe evacuation center in each ward. With the objective to orient the community on the local hazards and disaster preparedness, they are also planning to operate a DRR Learning Center. MEOC is also established in Changunarayan Municipality in B.S 2075 with support of Pragati Project. The municipality is also supported with a standard set of equipment like stockpiling of lifesaving equipment, IT equipment, and furniture along with SOP that outlines roles and responsibilities of municipality during emergency response. Pragati³ project also rendered sensitization and simulation exercises for strengthening institutional functionality with the community for enhanced disaster preparedness.

Though Palika is doing exemplary work on DRR, they are having setbacks for managing disaster risk and strengthening MEOC. Mr. Siddhant Neupane, a LEOC Focal Person, shares that the community is not aware of adverse consequences of disaster if they do not give priority to DRR. As LEOC Focal Person of Changunarayan Municipality, Mr. Neupane also recommended strengthening of LEOC, “Local elected bodies and municipal officials lack technical expertise on DRR. Therefore, Palika needs technical capacity building support from the federal government and DRR practitioners.” He further recommends “MEOC should be integrated with the Information Technology Section of the Municipality. This will help to operate and provide regular administrative services as well as DRR related service to locals.”

³ Promoting Resilient and Sustainable Urban Growth by Augmenting DRR Capacity of Local Authorities through Active Private Sector Involvement (PRAGATI) project aims to promote resilient and sustainable urban growth through DRR capacity building of local authorities and active involvement of the private sector and local population. It focuses on two urban Municipalities: Changunarayan and Shankharapur in Kathmandu Valley.

Annex 6: Leoc Presence Map



Map 2: LEOC Presence Map



Photo: HRRP Archive



Report by:



Primarily funded by:



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